

Extended depth of Field



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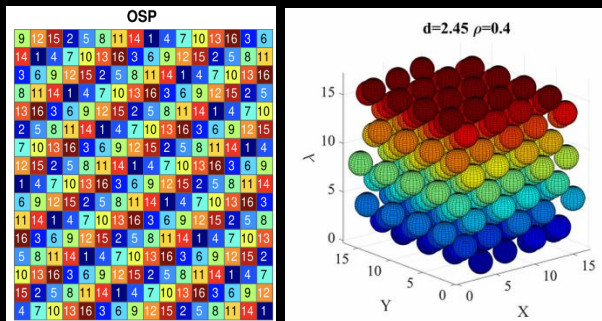


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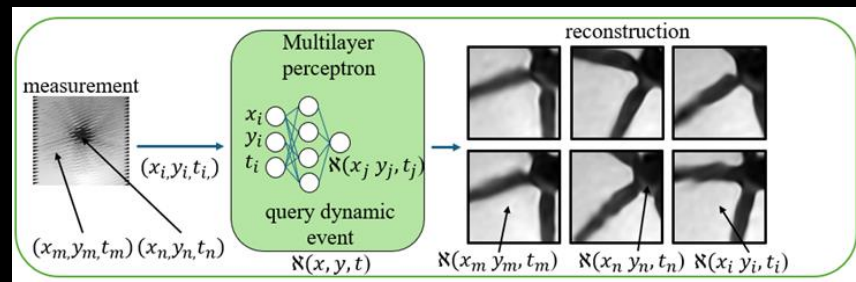
Computational Photography: Applications

- It can be applied to different signals
- It use compressive measurements
- To solve an inverse problem

Multispectral



Video

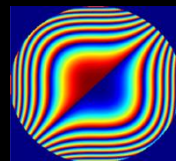


GT phase

Recovered phase

Recovered Amplitude

A woman

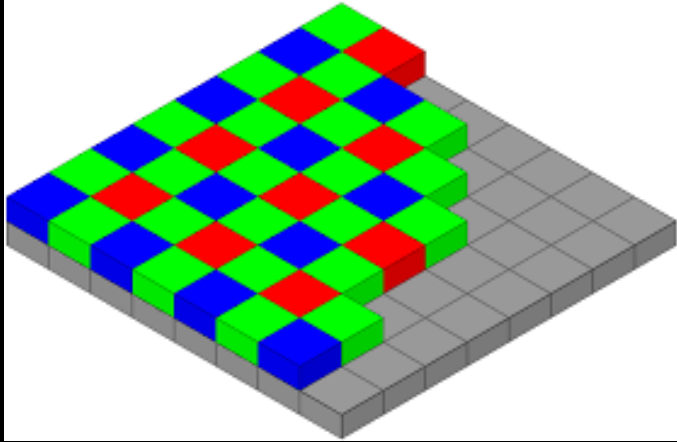


1995
Cubic Phase

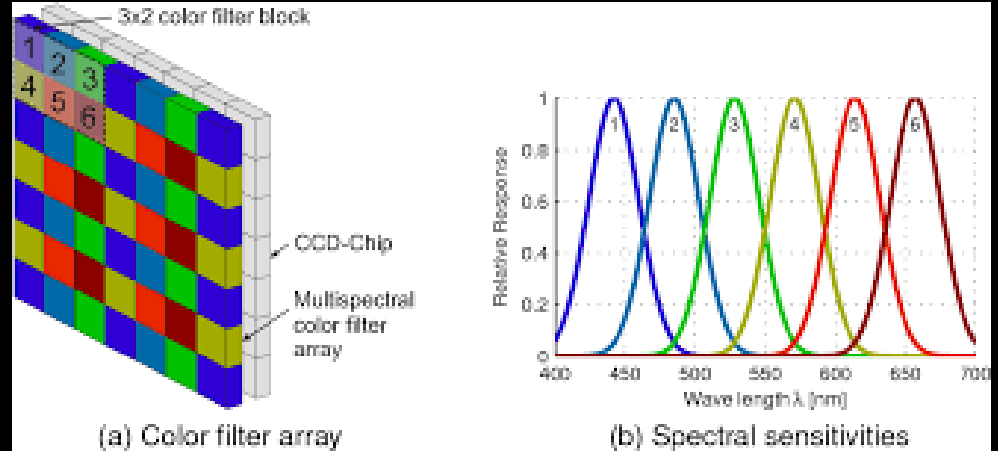


Depth

Multispectral Filter Array



Bayer filter



Multispectral Filter Array

The recovered signal is multispectral datacube

Random vs Sphere Packing

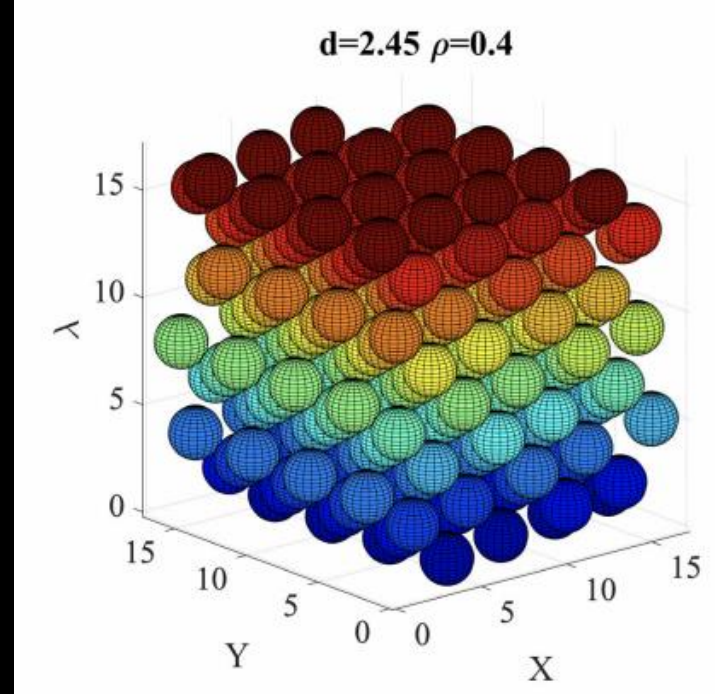
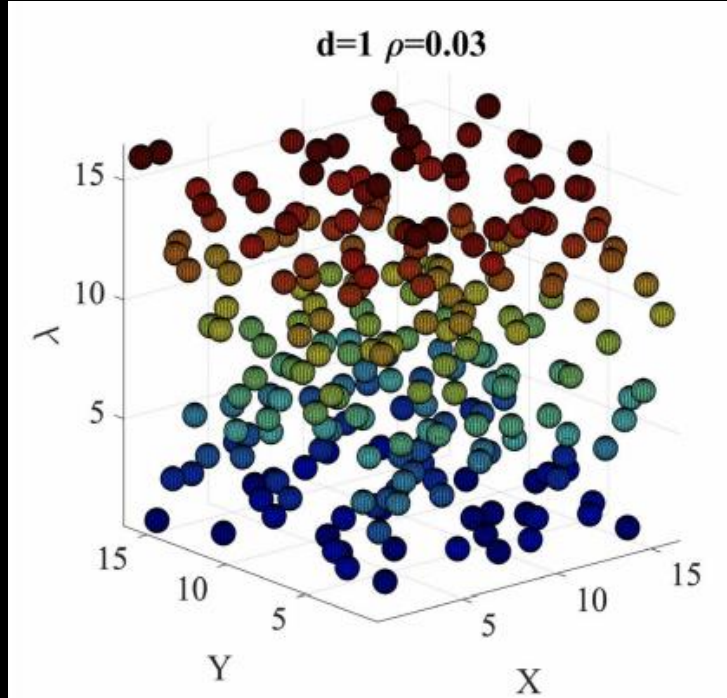
RND

4	5	2	9	8	11	13	7	14	6	10	15	12	3	1	16
5	11	14	13	2	10	9	15	6	8	7	1	4	3	12	16
16	7	1	6	14	8	11	13	3	2	15	4	10	9	5	12
16	9	5	4	10	7	1	6	12	8	3	2	15	11	13	14
9	7	13	15	10	3	12	2	1	16	14	6	8	4	5	11
15	2	4	11	13	1	16	6	14	5	10	9	7	12	8	3
14	7	11	9	6	4	3	2	15	8	13	16	10	12	5	1
7	15	5	11	10	14	13	4	8	9	12	16	3	2	6	1
2	6	1	5	9	11	3	4	16	8	13	10	14	7	12	15
7	15	10	12	1	9	16	11	3	8	6	13	14	5	2	4
6	12	2	14	7	4	1	15	11	5	16	10	13	8	3	9
11	8	12	14	15	7	1	10	13	4	3	9	16	2	5	6
5	2	13	3	15	1	9	6	7	12	8	14	10	11	16	4
6	9	2	15	7	13	16	12	10	4	1	8	11	14	3	5
7	1	12	13	16	3	2	8	14	11	6	15	5	9	4	10
10	11	6	12	13	14	2	9	1	15	4	16	3	7	5	8

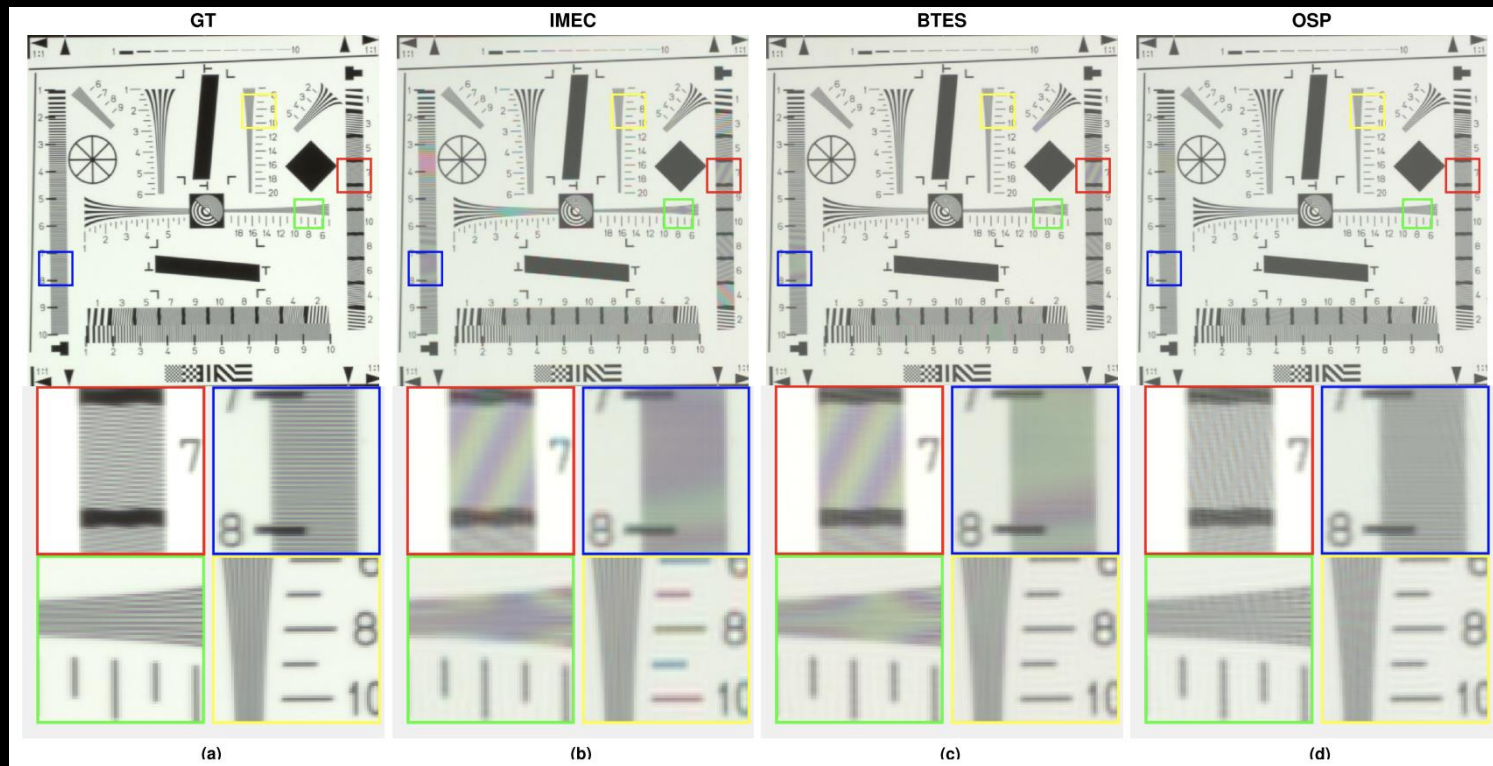
OSP

9	12	15	2	5	8	11	14	1	4	7	10	13	16	3	6
14	1	4	7	10	13	16	3	6	9	12	15	2	5	8	11
3	6	9	12	15	2	5	8	11	14	1	4	7	10	13	16
8	11	14	1	4	7	10	13	16	3	6	9	12	15	2	5
13	16	3	6	9	12	15	2	5	8	11	14	1	4	7	10
2	5	8	11	14	1	4	7	10	13	16	3	6	9	12	15
7	10	13	16	3	6	9	12	15	2	5	8	11	14	1	4
12	15	2	5	8	11	14	1	4	7	10	13	16	3	6	9
1	4	7	10	13	16	3	6	9	12	15	2	5	8	11	14
6	9	12	15	2	5	8	11	14	1	4	7	10	13	16	3
11	14	1	4	7	10	13	16	3	6	9	12	15	2	5	8
16	3	6	9	12	15	2	5	8	11	14	1	4	7	10	13
5	8	11	14	1	4	7	10	13	16	3	6	9	12	15	2
10	13	16	3	6	9	12	15	2	5	8	11	14	1	4	7
15	2	5	8	11	14	1	4	7	10	13	16	3	6	9	12
4	7	10	13	16	3	6	9	12	15	2	5	8	11	14	1

Sphere Packing



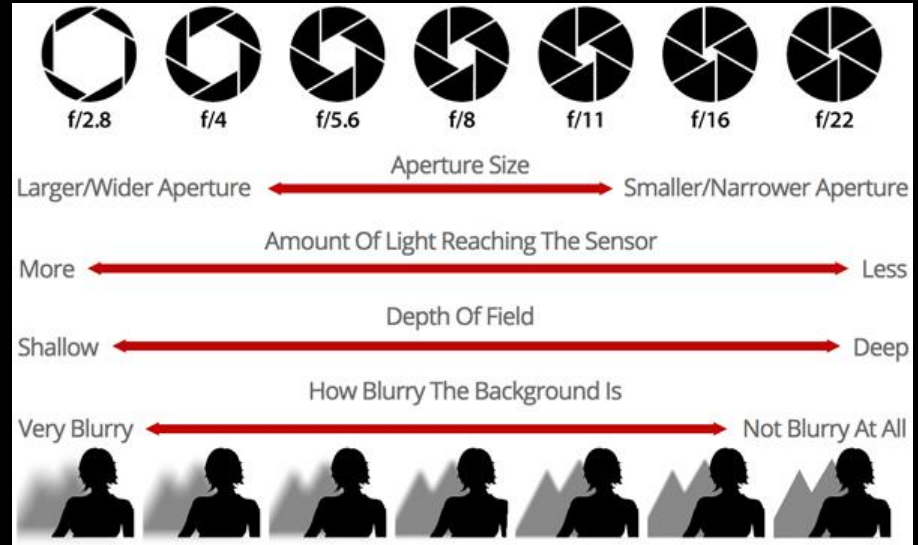
Reconstruction Results using MSFA



N. Diaz, A. Alvarado, P. Meza, F. Guzmán and E. Vera, "Multispectral Filter Array Design by Optimal Sphere Packing," in *IEEE Transactions on Image Processing*, vol. 32, pp. 3634-3649, 2023

Depth-of-Field (DoF)

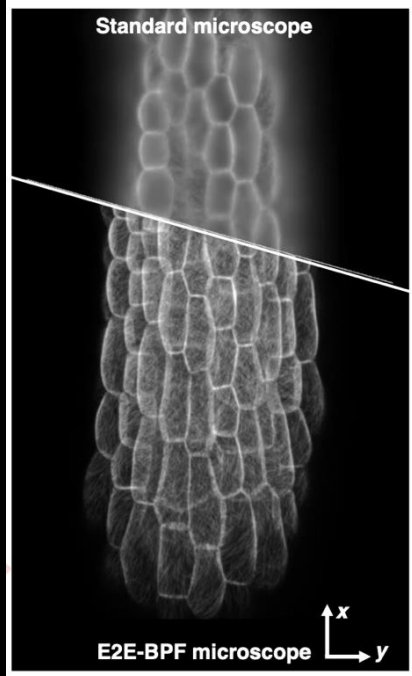
The **depth-of-the-field** refers to the depth range of an object that can be **sharply imaged** by a given optical imaging system and is determined by **operating wavelength**, **effective focal length** and **aperture size** of the imaging lens.



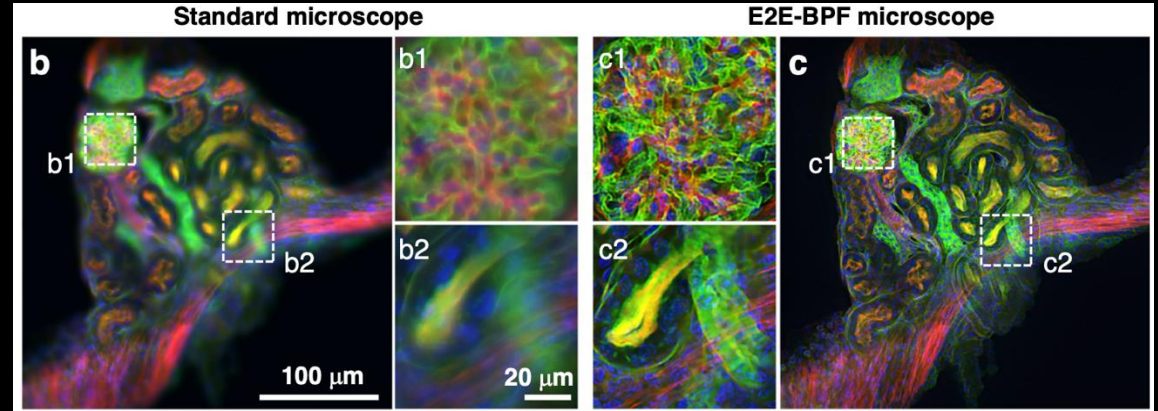
- Reducing the aperture size improves the depth of field
- But reduces the incoming light

Applications of EDoF

High-resolution imaging of large-scale objects without serial refocusing is challenging



E2E-BPF microscope¹

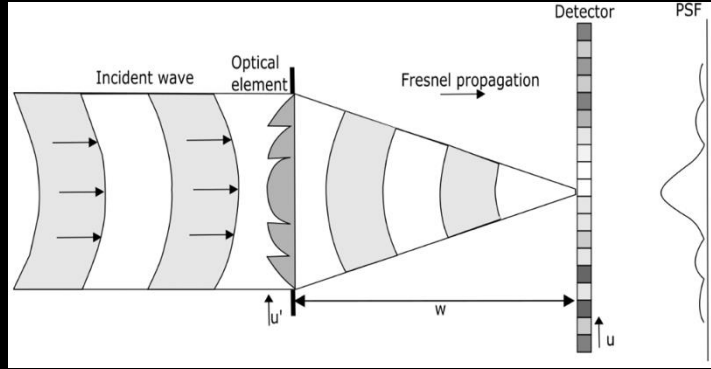


Traditional microscope vs EDoF microscope²

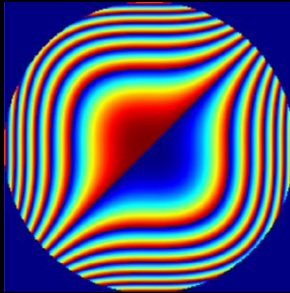
2. Seong, B., Kim, W., Kim, Y. et al. E2E-BPF microscope: extended depth-of-field microscopy using learning-based implementation of binary phase filter and image deconvolution. Light Sci Appl 12, 269 (2023).

Extended-DoF with Diffractive Optical Element

Two assumption when designing EDoF

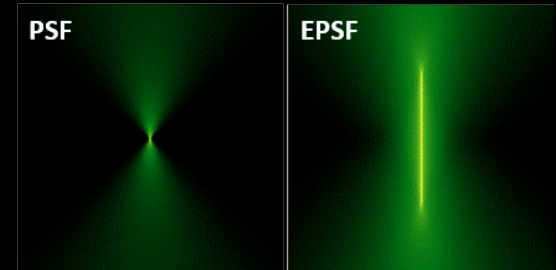


2. It is assumed that any resulting image will be an **intermediate image** and will require **digital processing**¹



1. The **incoherent optical system** is being modified by a rectangular separable **phase mask**¹

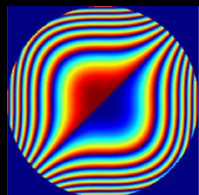
Standard vs Extended PSF



- Conventional cameras has narrow PSF
- The idea of EDoF is to promote PSF insensitive to misfocus

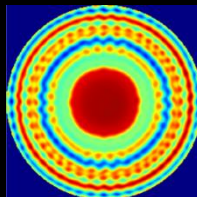
1. E. Dowski and W. Cathey, "Extended depth of field through wave-front coding," Appl. Opt. 34, 1859-1866 (1995).

Timeline of EDoF through Wavefront Coding



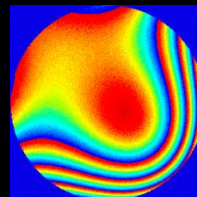
1995

Cubic Phase¹



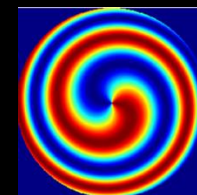
2021

End-to-end optimization³



2023

Hardware-in-the-loop⁴



2024

Helical Axicon

$$h_{\lambda}(s, t) = \text{mod}(\alpha\pi(s^3 + t^3), 2\pi), \\ |s| < 1, |t| < 1$$



f/8



f/1

$$d(s, t) = e^{(jL\theta_{(s,t)} + j\frac{2\pi}{\lambda}(f - \sqrt{f^2 - (s^2 + t^2)}))}$$

Helical **Axicon**

1. E. Dowski and W. Cathey, "Extended depth of field through wave-front coding," Appl. Opt. 34, 1859-1866 (1995).
3. Akpinar, E. Sahin, M. Meem, R. Menon and A. Gotchev, "Learning Wavefront Coding for Extended Depth of Field Imaging," in IEEE Transactions on Image Processing, vol. 30, pp. 3307-3320, 2021.
4. Pinilla, S., Fröch, J. E., Miri Rostami, S. R., Katkovnik, V., Shevkunov, I., Majumdar, A., & Egiazarian, K. (2023). Miniature color camera via flat hybrid meta-optics. Science Advances, 9(21), eadg7297.

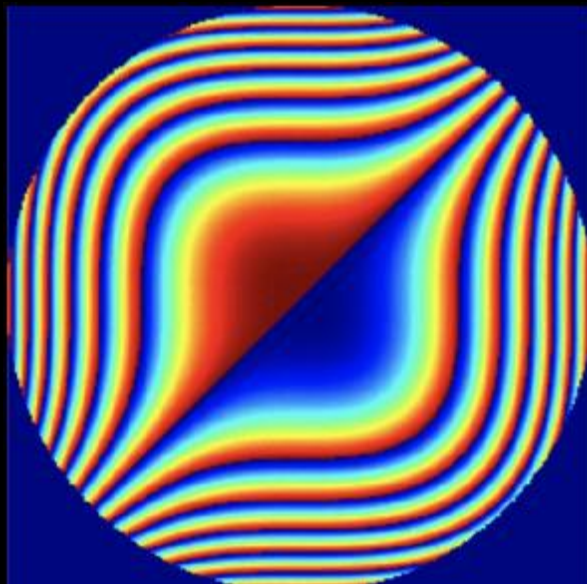
EDoF Discrete Model

$$Y_{RGB,z} = S_{RGB}(\textcolor{red}{PSF}_{\lambda,z} \odot X_{\lambda,z}) + \Omega_{\lambda,z}$$

designing the encoding is critical
to facilitate the inverse problem



measurement



DOE



Scene

Dowski-Cathey (1995)

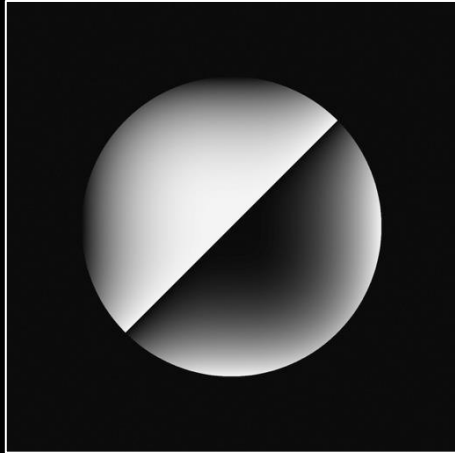
$$h_{\lambda}(s,t) = \text{mod}(\alpha\pi(s^3 + t^3), 2\pi), |s| < 1, |t| < 1$$

$\alpha=[6,16,63,126]$

$z = 0.5\text{m}$

$z = 1.0\text{m}$

$z = 2.0\text{m}$



Point spread function at different distances

Dowski-Cathey promotes PSF insensitive to misfocus¹

1.E. Dowski and W. Cathey, "Extended depth of field through wave-front coding," Appl. Opt. 34, 1859-1866 (1995).

Mathematical Model

$$PSF_{\lambda,z} = \left| \mathcal{F}\{Q_{\lambda,z}(s,t)\}_{\frac{x}{\lambda z_i}, \frac{y}{\lambda z_i}} \right|^2,$$

$\mathcal{F}\{.\}$ is the Fourier transform. Where the generalized pupil function is

$$Q_{\lambda,z}(s,t) = A(s,t)e^{\left(j\Phi_{\lambda}(s,t) + j\Psi_{\lambda,z}\left(\frac{s^2+t^2}{r^2}\right)\right)},$$

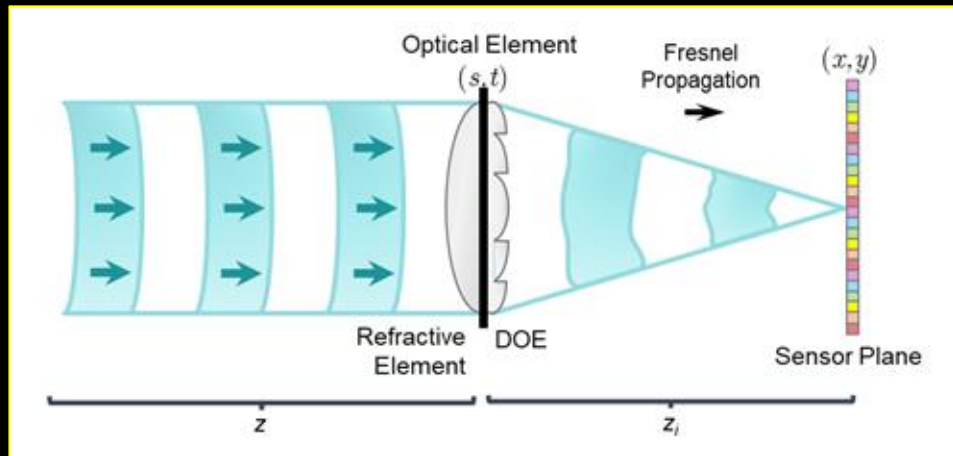
The thickness function of the diffractive element $\Phi_{\lambda}(s,t) = k(n_{\lambda} - 1)d(s,t)$,

$k=2\frac{\pi}{\lambda}$ is the wavenumber and n_{λ} is the wavelength-dependent refractive index.

Where the defocus coefficient is $\Psi_{\lambda,z} = \frac{\pi}{\lambda} \left(\frac{1}{z} + \frac{1}{z_i} - \frac{1}{f_{\lambda}} \right) r^2$

f_{λ} is the wavelength-dependent focal, r is the radius of the pupil, z distance of the object and z_i is the distance from the DOE to the RGB sensor.

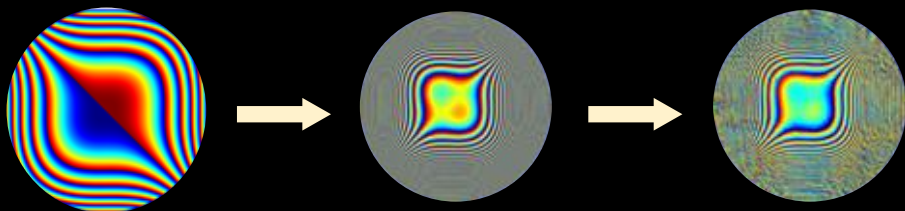
Diseño de Difractivos



Inicialización

NSGAIII

GD

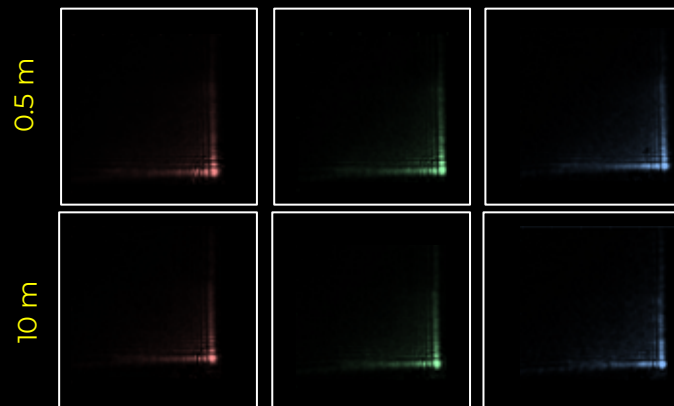


$$\Phi_{\lambda}^{\text{DOE}} = \Phi_{\lambda}^{\text{Cubic}} + \Phi_{\lambda}^{\text{Zernike}} + \Phi_{\lambda}^{\text{GD}}$$

Sistema Tradicional

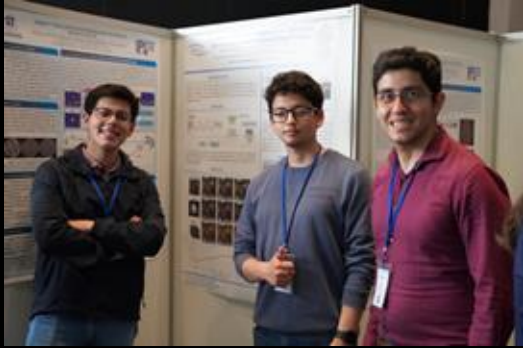


Sistema con Difractivo

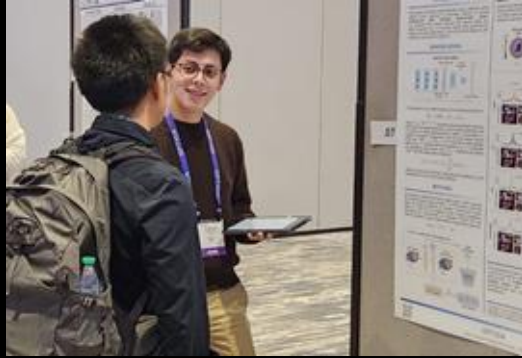


Conferencias y Pasantías

WACCI 2024,
Montevideo, Uruguay



Optica Imaging
Congress 2025,
Seattle, USA



Pasantía IPL UFRO
2025

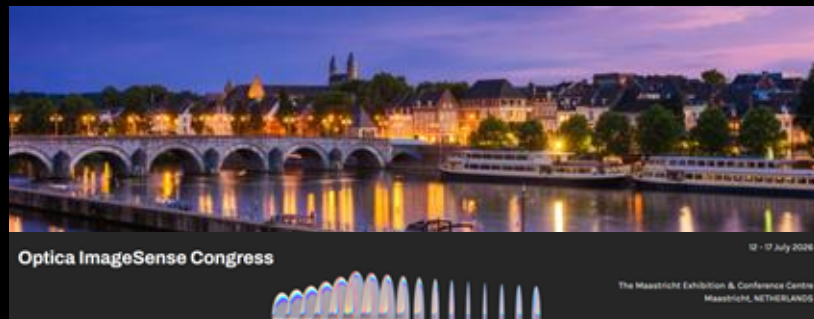


Conferencias Futuras

International Conference in Image Processing - ICIP 2026



Optica ImageSense Congress 2026



IEEE OJSP



OPTOLAB



Productivity: Conferences



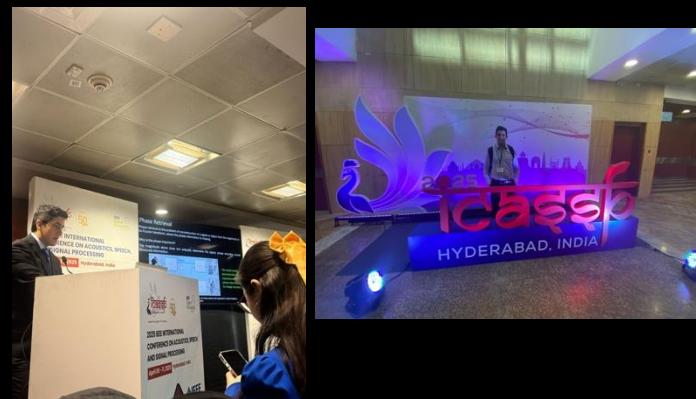
Optica imaging congress 2023, Boston, USA



2024, IEEE ICASSP, Seoul, Korea



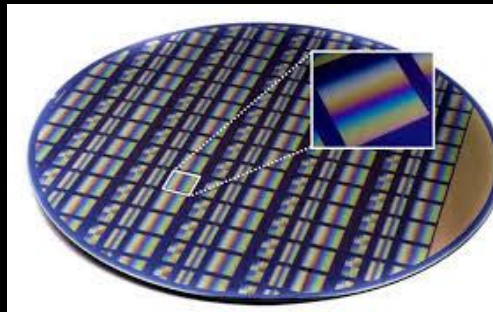
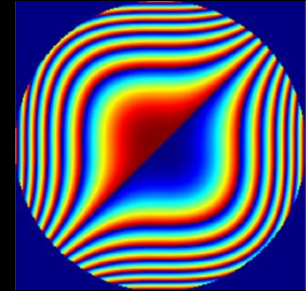
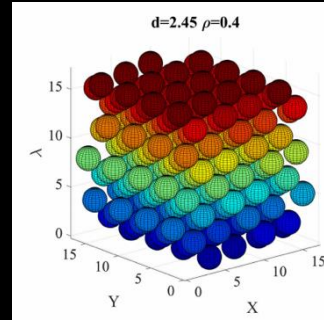
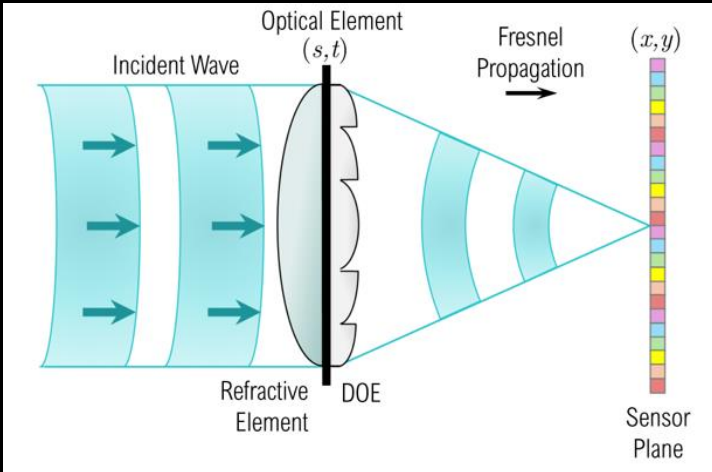
Optica imaging congress 2024, Toulouse, Francia



2025, IEEE ICASSP, Hyderabad, India

Fondecyt de Iniciación

Aim: to design an intelligent, chromatically invariant, encoded wavefront-sensing system for all-in-focus multispectral reconstruction (iCLEAR).



- To fabricate a multispectral camera
- To embed a multispectral filter array in camera
- To design a diffractive optical element

Research Network Worldwide

Chile



Esteban Vera
PUCV



Pablo Meza
Universidad de la
Frontera



Jorge Tapia
UTFSM



Dario Perez
PUCV

Asia



Felipe Guzmán
University of Tokyo

North America



Jinyan Liang
Université du
Québec, Canada



David Brady
University of Arizona,
USA

Europe



Samuel Pinilla,
Rutherford Appleton Lab,
United Kingdom



Karen Egiazarian,
Tampere University,
Finland



Jean Tourneret,
University of
Toulouse Francia



Marcus Carlsson
Lund University of Lund,
Sweden

Colombia



Henry Arguello, Universidad
Industrial de Santander,
Colombia⁹

Journal Productivity

IEEE Transactions on Image Processing (IF 13.7)

1. **N. Diaz**, A. Alvarado, P. Meza, F. Guzmán and E. Vera, “Multispectral Filter Array Design by Optimal Sphere Packing,” , vol. 32, pp. 3634-3649, 2023.
2. F. Guzman, **N. Diaz**, B. Romero and E. Vera, “Scalable Coding Strategy for High-resolution, High-compression ratio snapshot compressive video,” 2025.

Optics Express (IF 3.3)

1. **N. Diaz**, M Beniwal, M. Marquez, F. Guzman, C. Jiang, J. Liang, and E. Vera, “Single-mask sphere-packing with implicit neural representation reconstruction for ultrahigh-speed imaging,” , 2025.

Optics & Laser Technology (IF 5.0)

1. B. Romero, P. Scherz, **N. Diaz**, A. Cofré, J. Tapia and E. Peters, D Perez and E. Vera, “Phase Retrieval by Designed Hadamard Complementary Coded Apertures,” , pp. 1-13, 2025.

IEEE Open Journal of Signal Processing (IF 2.7)

1. E. Olivia, **N. Diaz**, P. Samuel, and E. Vera, “Multispectral extended depth-of-field imaging via stochastic wavefront optimization,” , pp. 1-13, 2025.

Investigador patrocinante: Dr. Esteban Vera
2023-2026 (**Fondecyt Postdoctorado**)

Áreas de investigación

- Applied Sphere Packing
- Single snapshot video
- Single Compressive video
- Multispectral Filter Demosaicking
- Phase retrieval with coded Illumination

